Quantifying and Attributing Polarization to Annotator Groups

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Abstract

Current annotation agreement metrics are not well-suited for inter-group analysis, are sensitive to group size imbalances and restricted to single-annotation settings, which render them insufficient for many subjective tasks such as toxicity and hate-speech detection. For this reason, we introduce a chance-corrected, dataset-level metric that attributes polarization to various annotator groups. Our metric enables direct comparisons between heavily imbalanced subgroups and between datasets and tasks, while also enabling analysis on multi-label settings. We apply this metric to three datasets on hate speech, and one on toxicity detection, discovering that: (1) Polarization is strongly and persistently attributed to annotator race, especially on the hate speech task. (2) Religious annotators do not fundamentally disagree between each other, but do with other annotators, a trend that is gradually diminished and then reversed with irreligious annotators. (3) Better educated annotators annotate with more nuance. Finally, we estimate the minimum number of annotators needed to obtain robust results, and provide an open-source python library that implements our metric.

Keywords: Polarization, annotation, annotators, socio-demographic, metric

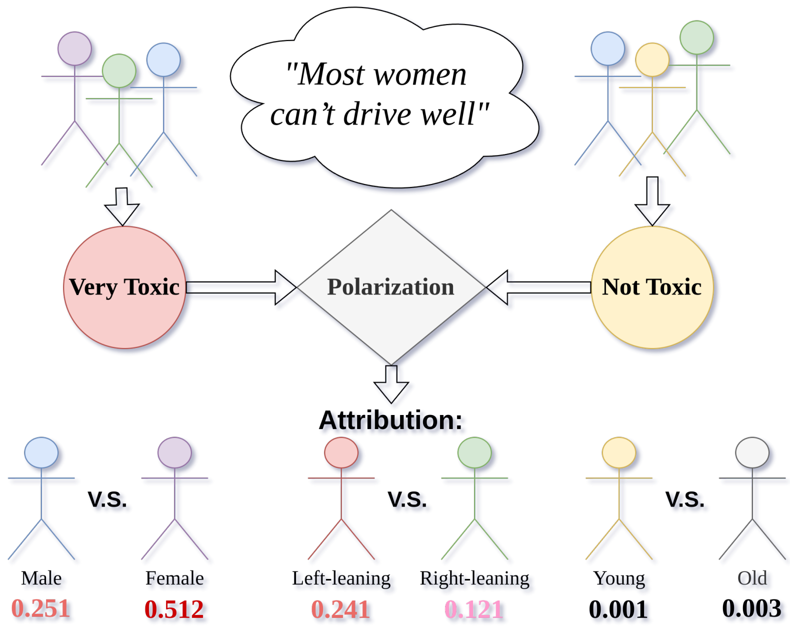

Fig. 1 The *apunim* framework attributes annotator polarization to various annotator subgroups.

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields with subjective judgements, such as content moderation, toxicity detection and hate speech detection. These tasks are especially important for training supervised learning models that remove or flag comments targeting vulnerable and minority groups in online spaces, where they are often vulnerable [1, 2]. However, annotations for such tasks are subjective and typically aggregated based on majority-voting. This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems [3].

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree with the majority. These fundamental disagreements are then ignored via aggregation (e.g., majority label voting) or filtering (e.g., keeping only items with large overall annotator agreement). Racist comments that are presented with coded language (usually referred to as "dog-whistles" [4]) are a common example. These comments may not be picked up by most annotators, but only by ones belonging in the targeted minority. Furthermore, majority-group annotators may themselves be biased against minority groups as was the case for toxicity models disproportionally marking African American speech as "toxic" [3].

Inter-annotator agreement is often used to detect such cases [5]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases, and fundamentally constrain analysis on single-label annotation. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [6, 7]. While we can detect polarization [6, 7], there is little work on how to detect whether it is actually caused by marginalized groups. An existing mechanism operating on individual comments [7], operates only as a binary classifier, and as such does not produce quantifiable/comparable results. It also only tests for the extreme case where *all* present polarization can be attributed to any single annotator group, which is rare in real-life datasets (§4).

We instead propose the *apunim ("Aposteriori Unimodality") metric*. Unlike previous methods, *apunim* is designed to detect any level of attribution for each annotator group, providing a more fine-grained attribution. Additionally, unlike previous approaches which only work on a comment-to-comment basis, *apunim* generalizes results for all annotated items, which is particularly important for our formulation due to the limited number of annotations per item. By considering the entire set of items, *apunim* helps to identify genuine polarization and distinguish it from chance attribution, which is essential when analyzing large datasets. Our contributions are:

1. We introduce a new metric that quantifies polarization, which can be attributed to specific annotator groups, and we develop a parametric test for statistical significance. This metric works for both single-label and multi-label annotation setups, is generalizable across any domain and modality (e.g., video/image classification, toxicity/hate-speech detection) and enables quantitative comparisons across datasets and dimensions (e.g., gender, race, political affiliation). Unlike previous work, our metric is designed to apply to entire datasets instead of individual items.
2. We discover issues not yet acknowledged in literature. We are the first to report the existence of *inherent polarization*: most annotated items having some degree of polarization that can not be attributed to *any possible partition* of annotators. We also find that annotator groups may exhibit opposing polarization attribution between items, which can cancel each other out when not handled properly. Our proposed metric is designed to avoid these issues.
3. We apply our metric to four Natural Language Processing (NLP) datasets focused on Toxicity and Hate Speech detection. We find that annotator race/ethnicity contributes to polarization across all datasets, but more so on Hate Speech tasks. We also discover that educated annotators are generally more nuanced, and that in some cases, polarization is either concentrated around similar clusters of groups (e.g., age and education) or that annotators become more polarized the more extreme their opinions become (e.g., religion).

We release an open source Python library that implements the calculation of the metric and its statistical significance, which can be directly installed via PyPi (see Appendix C).[1] Furthermore, we provide an estimation on the minimum number of annotators required for robust estimates, and we release the code for the experiments, graph creation, and analysis on our project's repository.[2]

[1] Library: https://pypi.org/project/apunim Documentation: https://dimits-ts.github.io/apunim
[2] Replication code: https://github.com/dimits-ts/aposteriori-unimodality

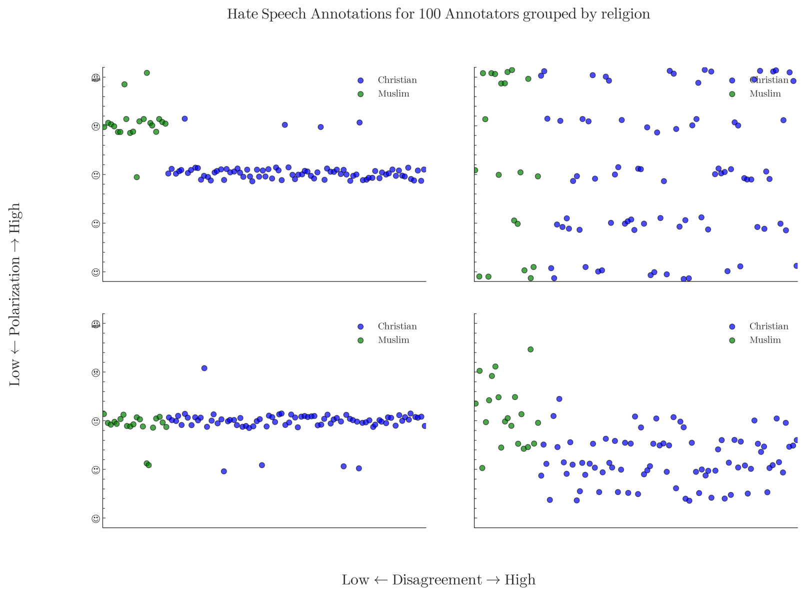

Fig. 2 Disagreement is similar to variance, overlooking cases where annotators may be "split down the middle". Polarization on the other hand, identifies multiple clusters even if they are close together.

2 Related Work

Annotator agreement

There are many popular metrics for measuring inter-annotator agreement such as Cohen's kappa, Krippendorff's alpha and Fleiss's kappa. All of these metrics attempt to distinguish genuine disagreement among annotators from noise (or "random disagreement") using statistical methods. However, a number of them can not be generally used for comparing intergroup annotation agreement without extending them (i.e., computing pairwise Cohen's kappa for all groups), or placing constraints on when they can be used (i.e., no missing labels and same number of annotations per annotator for Fleiss's kappa). They have also been criticized for ignoring fundamental differences that can arise from different social, cultural and demographic backgrounds [8, 9], and for their inability to quantify and compare results across experiments [10].

Recent work has attempted to work around these limitations, for example by adapting Cohen's kappa for subgroup agreement [10] or creating a Bayesian model to explicitly differentiate between subgroup disagreement and noise [11]. While these extensions do improve the performance of these metrics, they are typically less explainable and intuitive. They are also constrained by the same basic assumptions of disagreement metrics, such as not being suitable for multi-label annotations (where annotators can pick more than one option for each item).

Annotator polarization

Due to their nature, disagreement metrics may not perform well when comparing annotator groups with greatly mismatched sizes [8, 9]. Figure 2 shows an example of an annotation task where the minority group annotators may be ignored when computing disagreement, due to the overwhelming number of majority-group annotations. Polarization [7] attempts to solve this problem by framing disagreement as a cluster identification problem, where similar annotations will belong in a single cluster, disregarding its size. This enables metrics based on polarization to work with multi-label annotations, since we are no longer comparing individual annotations against those of other annotators, but rather each annotator's histograms (see §3.1).

Attributing polarization to subgroups is a relatively new idea [7]. Under the name *aposteriori unimodality*, the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, this has to mean that those subgroups are "causing" the exhibited polarization. Though promising, this idea is not practically usable, since subgroup polarization being exactly zero is an edge-case in real-life annotation tasks (§4). Additionally, getting polarization estimates from realistically few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarising) instead of a quantifiable value (i.e., when presented with noisy estimates, we would ideally distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49 on a scale of [0, 1]). As was shown in [12], such a binary analysis can be applied to discover whether Large Language Models (LLMs) are biased in the same way as specific human subgroups. Also, ordinal variables can extract "better" categorical labels via dataset-dependent thresholding, an idea that is promising yet limited (see §5.3). Overall, however, this work showed that polarization is a versatile tool, but also displays the constraints of using the non-quantifiable binary classification given by the traditional aposteriori unimodality framework. For instance, the authors have to resort to Cohen's kappa agreement for any kind of comparison between LLMs and human annotator groups, which downgrades polarization to the role of filtering texts.

3 How to attribute polarization

3.1 Problem Formulation

Annotator groups

Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to split annotators according to their Personal Characteristics (PCs). Each of these characteristics is a dimension Θ composed of multiple groups θ_1, θ_2, $\ldots$; e.g. if we are investigating gender, then $\Theta = \{\theta_{male}, \theta_{female}, \theta_{nonbinary}\}$. In previous publications [3, 7, 13, 14], these characteristics were exclusively sociodemographic in nature, although recent research [12, 15] confirms that personal opinions and experiences shape annotator decisions as well.

Annotations

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated items.[3] These items could be of any modality, such as comments in a discussion, images, videos or survey

[3] We use the general term "item" instead of domain-specific terms, even though the examples presented in the paper are comments. The mathematical term "point" is equivalent.

questions. We assume that how polarizing an item is depends on two variables: *inherent ("apriori") polarization*, and polarization caused by *different annotator PCs ("aposteriori" polarization)*. Each item c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, \theta_1), (a_2, \theta_2), \ldots\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task grouped by gender, would be formulated as:

$$A(\text{``Could be better, could be worse''}) = \{(+, F), (-, M), (-, F), \ldots\}$$

Polarization

In the absence of polarization, annotation distributions are typically unimodal: annotators may disagree on finer details—shown as a bell-shaped distribution around a central mode—rather than on fundamentally different interpretations, which would instead give rise to multimodal distributions. To our knowledge, the only polarization metric explicitly grounded in this assumption is the normalized Distance From Unimodality (nDFU) [7]. The metric takes values in the [0, 1] range, with $nDFU \approx 0$ indicating no polarization (unimodal distributions) and $nDFU \approx 1$ corresponding to complete polarization (highly multimodal distributions). Our objective is therefore to quantify the extent to which the ($nDFU$) score of a polarizing item can be (partially) explained by a given PC group θ.

3.2 Intuition

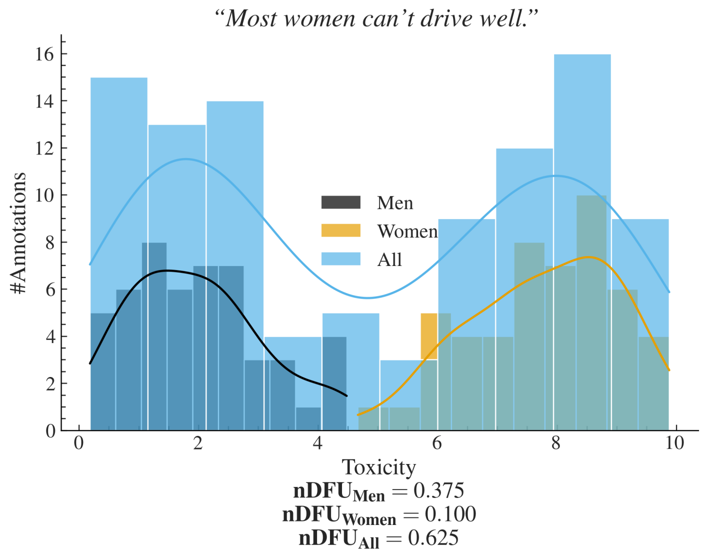

Fig. 3 Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

Polarization attribution

Intuitively, a group $\theta_i \in \Theta$ partially explains the polarization of an item c when the annotations of that group $A(c \,|\, \theta_i)$ exhibit higher polarization compared to the full set of annotations $A(c)$, where $A(c \mid \theta_i) = \{a(c;\theta) \in A(c) \mid \theta = \theta_i\}$. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators. The annotations are generally polarized ($nDFU_{all} = 0.62$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.10$), since most agree that the comment is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.37$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

Previous work [7] stopped on item-to-item (comment-to-comment) analysis. In practice however, this approach is inherently noisy due to a lack of information (i.e., limited annotations) relating to each individual item. In order to get meaningful information, we need to consider the annotations from all items, as the number of items is usually several orders of magnitude larger than the number of annotations per item (§4). We thus reformulate our hypothesis; if an annotator subgroup causes some of the underlying polarization in our dataset, we expect that this attribution will persist across multiple polarizing items.

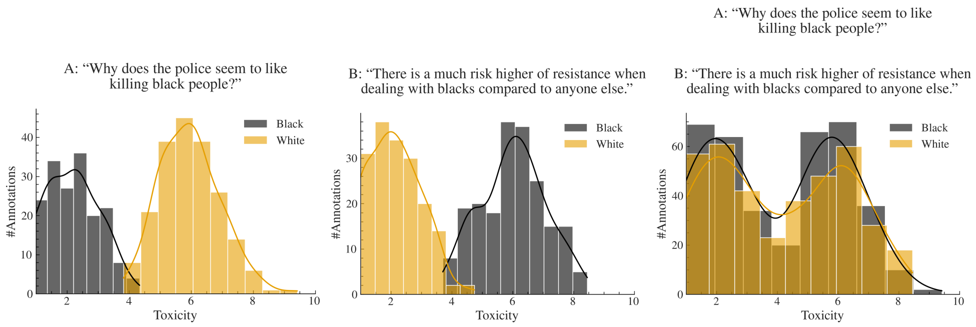

Fig. 4 Example of a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments (right-most image), we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations) as shown in the center and left-most images, we end up comparing two bimodal distributions formed by combining both comments within each group. This aggregation can hide where the polarization actually comes from.

Dataset polarization

Given this observation, we would be tempted to quantify dataset-level polarization attribution by simply aggregating all annotations in the dataset and comparing $A(d)$ with $A(d \,|\, \theta)$.[4] In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each item may be annotated

[4]Slight abuse of notation; we are using all annotations in dataset d as if belonging to the same item.

by only a handful of annotators (5 is a common number in many publications), leading to a best-case even split between groups (for 5 annotators, a best case scenario would be only two groups with a 3-2 split). Since nDFU is at its core a histogram-based estimation, three annotations may not be enough to acquire a reliable histogram, and thus estimate of polarization. This is discussed in more depth in §6.

However, arbitrarily combining annotations would not work well in practice. Figure 4 illustrates a hypothetical discussion with two comments, both of which are toxic, but where the two groups of annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that item-level polarization attribution has a *direction*, when viewed from distance, which is not captured by nDFU.

3.3 Aposteriori polarization

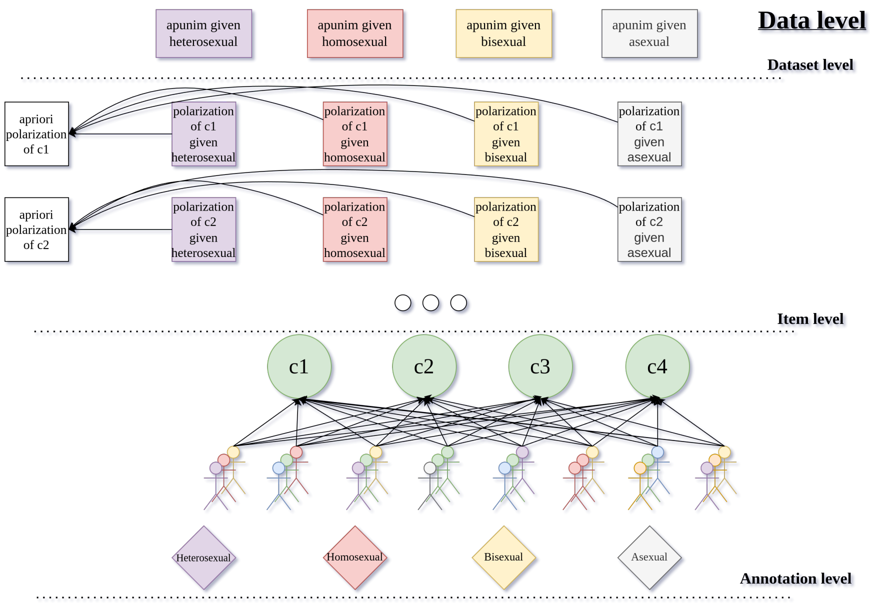

Fig. 5 An overview of how annotation information is sequentially aggregated for the *apunim* metric. For each item we calculate the apriori polarization and the sample aposteriori polarization given each PC group. We then aggregate both apriori and aposteriori polarization and compare them on the dataset level to obtain the polarization attribution ("apunim") values for each group.

We do not directly compare the differences in nDFU between $A(c)$ and $A(c \mid \theta)$, as in §3.2, to avoid comparing statistics between a subset and its superset. Standard practise dictates that we should compare the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this does not work in the (common) case where we are studying a PC with two subgroups.[5] If we applied the subset-vs-complement method in Figure 3 where we consider the gender dimension, we would end up subtracting the nDFUs of men and women, which are extremely similar, thus concluding that there is no polarization present.

3.3.1 Apriori Polarization

In a scenario in which each item was annotated by many annotators, we could use sampling or bootstrapping to compare apriori and aposteriori polarization per comment. This, however, is an unrealistic assumption, since the costs of human annotation force researchers to use few annotators for each task [16]. To get around this limitation, we create random stratified splitswhich effectively "breaks up" the effect of any single isolated group on the polarization of the sample.[6] The randomly partitioned annotations are then sampled t times. Formally, the (average) apriori polarization for a single item c is then given by:

$$P_{apr}(c) = \frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c))) \tag{1}$$

where $\tilde{r}$ is the random partition operator with group sizes matching the group sizes of PC Θ. Unlike $A(c)$, we can use this formulation for direct comparisons with $A(c \mid \theta)$.

3.3.2 Dataset Polarization: Apriori and Observed

While the low number of annotators is an important limitation when acquiring the observed polarization [16], this is decisively not the case with the number of items themselves which can range up to hundreds of thousands. Thus, we have the luxury of filtering them before computing our metric, as proposed in previous work [12]. Since our metric attributes polarization, unpolarizing items and those composed of only a single annotator group are useless. We define the subset S_d of dataset d such that:

$$S_d = \{c \in d \mid nDFU(c) > \alpha; \Theta^c > 1\}$$

where α is a small positive number and Θ^c is the number of distinct annotator groups in $A(c)$. This definition filters out a large fraction of unpolarized items (§4), reducing noise and significantly lowering computational costs (see Appendix D). Then, we can estimate how much polarization is generally present in the dataset, irrespective of any PC, by simply averaging the *apriori* polarization of its items:

$$P^d_{apr} = \frac{1}{|S_d|}\sum_{c \in S_d} P_{apr}(c) \tag{2}$$

[5] Even when there are more than two groups, we may end up with no items that were annotated with more than one annotator of the other minority groups, meaning that we can not get a histogram estimation. This occurred frequently in datasets with low numbers of comments such as the Sap dataset (§5).

[6] E.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20.

Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by factors of the investigated Θ dimension, because polarization insists disregarding that dimension. Similarly to Equation 2, we also define polarization that was *observed* by a specific group of annotators, e.g., θ, as:

$$P^d_{obs}(\theta) = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(A(c \mid \theta)) \tag{3}$$

This formulation implicitly assumes that dataset polarization can be estimated by averaging polarization scores across its items. We consider this assumption reasonable, because the mean is a simple and intuitive operator, exhibiting appropriate sensitivity to outliers. Given that polarization values lie in the [0, 1] range, a single item with extreme polarization is unlikely to substantially affect the estimate of overall polarization. In contrast, the presence of many such extreme values will meaningfully increase this estimate, which is the pattern we want to capture.

3.3.3 Apunim

We can now compare the polarization that is attributed to a single-dimensional annotator group (Equation 3) to a prior one (Eq. 2). We define our normalised metric, which we call as *"apunim" (Aposteriori Unimodality)*, as:

$$apunim(\theta) = \frac{P^d_{obs}(\theta) - P^d_{apr}}{1 - P^d_{apr}} \tag{4}$$

The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 5. Since $nDFU \to [0,1] \Rightarrow apunim \to [-1,1]$. The interpretation of this metric boils down to three scenarios:

- If $apunim(\theta) \approx 0$, the exhibited polarization among annotators of group θ does not meaningfully differ from all other groups. In other words, it is indistinguishable from noise (represented by the apriori polarization).
- If $apunim(\theta) > 0$, then the polarization observed in the annotation task rises when this group is added to the annotation pool (equivalently, intra-group polarization is lower than apriori polarization). For example, if we observe $apunim(\textit{"African American"}) = 0.3$, $apunim(\textit{"White"}) = 0.001$ in a hate-speech detection task, it is likely that African American annotators fundamentally disagree with white annotators on what constitutes racism, regarding dataset d.
- Conversely, if $apunim(\theta) < 0$, then the overall polarization *drops* when we add annotators from this group to this task. Mathematically, this occurs when intra-group polarization is higher than apriori polarization. This may indicate that *other dimensions (acting as confounding variables) are exerting heavy influence on the sample* or that annotators of that group may be very polarized between themselves. For example, if we observe $apunim(Latino) = -0.3$, $apunim(White) = 0.001$ in a hate-speech detection task, it is likely that Latino annotators fundamentally disagree with *each other* (compared to African American or white annotators) on which texts in the dataset d constitutes racism.

Algorithm 1 P-value estimation for aposteriori polarization

Require: Dataset d, annotator groups Θ
Require: Filtered items S_d
Require: Number of random partitions t
Ensure: p-value for testing whether $apunim(\theta) \neq 0$

$P^d_{obs}(\theta) = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(A(c \mid \theta))$ ▷ As in Equation 3
$P^d_{apr} = \frac{1}{|S_d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c)))$ ▷ As in Equation 2
$apunim = \frac{P^d_{obs}(\theta) - P^d_{apr}}{1 - P^d_{apr}}$ ▷ As in Equation 4
$rand\_apunims = \{\}$
for $i = 1, 2, \ldots, t$ **do**
    $rand\_obs = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(\tilde{r}_i(A(c))$ ▷ Single random partition
    $rand\_apriori = \frac{1}{|S_d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c)))$ ▷ Same as P^d_{apr}
    $rand\_apunim = \frac{rand\_obs - rand\_apriori}{1 - rand\_apriori}$
    $rand\_apunims = rand\_apunims \cup rand\_apunim$
end for
$p = Student\text{-}T(rand\_apunims \neq apunim)$
return p

3.4 P-value estimation

The strict condition $apunim(\theta) \approx 0$ is not sufficient due to inherent annotation variance. We initially formulated the following statistical hypothesis:

$$H_0 : \forall \theta \in \Theta,\ apunim(\theta) = 0 \text{ and } H_1 : apunim(\theta) \neq 0$$

However during early experimentation, we observed that *apunim* values were rarely statistically close to 0, which we hypothesize to be caused by the presence of inherent polarization (see §3.3). This would indicate that this formulation is not strong enough, and leads to frequent Type I errors.

For this reason, we reformulate our hypothesis; If the apunim value is statistically significant, we would expect it to significantly differ *from apunim values exhibited by randomized groups* on the same sample. We use the individual randomized partitions $r_i(A(c))$ to construct pseudo-apunim values which are then compared to the actual apunim value of the sample using a Student-T test, as shown in Algorithm 1. Thus, we test a much stricter hypothesis:

$$H_0 : \forall \theta \in \Theta,\ apunim(\theta) = \bar{apunim}(d),\ H_i : apunim(d, \theta) \neq \bar{apunim}(d),\ \forall \theta \in \Theta$$

where $\bar{apunim}(d)$ denotes the mean apunim values of each random partition $\tilde{r}_i(A(c))$.

Our test assumes that there exist (1) enough data-points to reliably run the means-test, and (2) enough annotations in each and PC group to estimate the nDFUs in Equations 3,2. The former can be safely assumed for most datasets, because they feature enough data-points to reliably assume normal residuals (see §4), which is why we use the Student-T test [17]. We discuss the latter assumption in §6.

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [18]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [19]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

4 Datasets

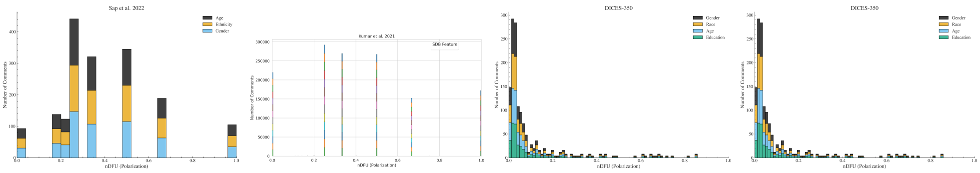

Fig. 6 Histogram of dataset polarization for the human datasets considered in this paper per PC.

Table 1 Overview of the datasets used in this study. *# Annotators* refers to the number of annotations per comment. For detailed information on the annotation counts for the comments of each dataset, see Appendix B.

Dataset	#Comments	#Annotators	#PC dims	Task
Kumar [15]	106,035	5	10	Toxicity
Sap [13]	626	5–6	3	Hate Speech
DICES-350 [14]	72,103	70–76	4	Hate Speech
DICES-990 [14]	43,050	123	4	Hate Speech

We use four datasets from current NLP research that include PC information at the level of individual annotators.[7] Specifically, we use the “Kumar” [15] and “Sap” [13] datasets, which are concerned with Toxicity and Hate Speech detection respectively, as well as the DICES-350 and DICES-990 datasets [14], which focus on LLM response safety. For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to Hate Speech detection. While the Toxicity and Hate Speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Thus, we expect the sources of annotator disagreement and bias that arise in these tasks will be broadly similar.

The datasets differ substantially in scale and annotation design. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC and personal-experience dimensions. In contrast, the

[7]While we would like to test our framework on other research areas and domains, NLP remains the only large field to our knowledge that systematically investigates annotator biases and includes the necessary PC information.

DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. A brief overview of dataset characteristics is provided in Table 1. Across all datasets, we observe a non-trivial degree of polarization in most annotations, as illustrated in Figure 6.

5 Results

5.1 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the toxicity/racism annotations. The results for the DICES-350 and DICES-990 datasets can be found in Tables 2 and 3, while Tables 4 and 5, display the results for the Kumar and Sap datasets respectively. We sample 20,000 comments from the Kumar dataset due to computational costs (see Appendix D). Note that some groups are not included, since there were no polarized items annotated by more than one annotator of these groups. For details on other parameters and execution times, refer to Appendix D.

5.2 Which factors contribute to polarization?

Table 2 Aposteriori unimodality results for the DICES-350 dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	0.0186	9703
	2) Millenial	-0.0059	11268
	3) Gen. Z	-0.0066	17528
Education	College or higher	0.0124	23475
	High school or lower	-0.0152	12833
	Other	-0.0214*	2191
Gender	Man	-0.0193	19093
	Woman	0.0215	19406
Race	African Am.	-0.0428***	9077
	Asian	0.0659***	8138
	Latino	-0.0030	6886
	Multiracial	-0.0001	5008
	White	-0.0198	9390

Table 5: Aposteriori unimodality results for the Kumar dataset.

PC dimension	Value	apunim	support
Age	1) 18 - 24	-0.0090	9742
	2) 25 - 34	0.0324***	33203

Continued on next page

Table 5: Aposteriori unimodality results for the kumar dataset.

PC dimension	Value	apunim	support
	3) 35 - 44	-0.0093	21055
	4) 45 - 54	0.0006	11025
	5) 55 - 64	-0.0037	6123
	6) 65 or older	0.0001	2567
Education	1) Doctoral degree	0.0048	1019
	2) Professional degree	0.0038	1321
	3) Master's degree	0.0147**	13420
	4) Bachelor's degree	0.0401***	34085
	5) Associate degree	-0.0113**	9122
	6) College, no degree	-0.0335***	16666
	7) High School graduate	-0.0079*	7326
	8) No high school	0.0004	462
Ethnicity	Asian	-0.0019	5062
	Black	0.0110*	10999
	Hispanic	-0.0025	2389
	Multiracial	-0.0041	3682
	Other	0.0022	1507
	Unknown	-0.0030	1160
	White	-0.0377***	45306
Gender	Female	-0.0278***	40686
	Male	0.0275***	37929
	Nonbinary	-0.0008	489
	Unknown	-0.0029	621
Has Been Targeted	False	-0.0633***	45234
	True	0.0349***	24491
Is Parent	No	-0.0707***	37231
	Prefer not to say	-0.0023	831
	Yes	0.0535***	41493
Is Transgender	No	-0.0894***	11978
	Prefer not to say	0.0005	856
	Yes	0.0013	2306
Political Affiliation	Conservative	0.0331***	23010
	Independent	-0.0157***	22818
	Liberal	-0.0287***	32917
	Other	-0.0049	1822
	Prefer not to say	0.0015	3548
Religion Important	1) No	-0.0804***	26008
	2) Not very	-0.0154***	10055
	3) Somewhat	0.0210***	20169
	4) Very	0.0487***	27284
	Prefer not to say	0.0001	1179
Seen Toxicity	False	0.0311***	20185
	True	-0.0717***	43265
Sexual Orientation	Bisexual	0.0175***	10005
	Heterosexual	-0.0571***	38048
	Homosexual	-0.0017	2975
	Other	-0.0003	728
	Prefer not to say	-0.0017	1751
Technology Impact	1) Very negative	-0.0009	784
	2) Somewhat negative	-0.0080	7973
	3) Neutral	0.0027	10720
	4) Somewhat positive	-0.0265***	40477
	5) Very positive	0.0208***	22471
Toxicity Problem	1) Never	0.0058	4962
	2) Rarely	0.0177***	14457
	3) Occasionally	-0.0058	29476
	4) Frequently	-0.0217***	24774
	5) Very Frequently	-0.0145***	10656

Table 3 Aposteriori unimodality results for the DICES-990 dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	-0.0471***	14384
	2) Millenial	-0.0043	38436
	3) Gen. Z	0.0644***	13741
Education	College or higher	0.0095	59142
	High school or lower	-0.0751***	7419
Gender	Man	0.0278***	32494
	Woman	-0.0235***	33471
Race	African Am.	0.0909***	5810
	Asian	0.0100	18428
	Latino	-0.1328***	6610
	Other	0.0753***	24321
	White	-0.1196***	11392

Table 4 Aposteriori unimodality results for the Sap dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	0.0192	743
	2) Millennial	0.0189	1792
	3) Gen. Z	-0.0153	239
Ethnicity	Black	-0.2967***	1041
	White	0.1400***	2048
Gender	Male	0.0942***	1635
	Female	-0.1810***	1333

Polarization is consistently attributed to annotator Ethnicity across all datasets, and especially so on hate-speech detection tasks

Across all four datasets, we observe a consistent pattern in which annotation polarization can be attributed to differences in annotator ethnicity—in many datasets this is the strongest contributor to polarization. In each case, statistically significant effects emerge for at least one of the major ethnic groups represented in the data—most notably *White* and *African American* annotators. Additionally, polarization contribution is substantially more pronounced in the Hate Speech datasets than in the Toxicity dataset. This suggests that *ethnic background constitutes a systematic source of annotator disagreement* rather than isolated noise, and is *more prevalent on tasks explicitly investigating racism.*

Some dimensions exert an overall influence on the dataset

Some PC dimensions do not necessarily sum to zero, which suggests that *these dimensions may exert a systematic influence* on the total polarization of the dataset. In other words, some annotator characteristics can consistently increase or decrease observed polarization (§3.3). In the Kumar dataset, for example, respondents who report having been previously targeted contribute positively to polarization ($apunim = 0.0349$), while those who have not been targeted contribute much more negatively ($apunim = -0.0633$). Similarly, transgender annotators contribute negatively to polarization ($apunim = -0.0894$) which is not balanced out by cisgender ones ($apunim = 0.0013$). These asymmetries indicate that some characteristics shape the overall polarization of the dataset rather than simply redistributing it across groups.

Educated annotators tend to increase polarization

Education tends to have a moderate but consistent directional effect across all three datasets where it is present. The direction of this effect is consistent; the more educated an annotator is, the more polarized their ratings become. In the Kumar dataset, for instance, Bachelor's-degree holders show a substantial positive contribution ($apunim = 0.0401$), while groups with lower education such as "College, no degree" exhibit a negative contribution ($apunim = -0.0335$). This may indicate that educated annotators are more likely to exhibit subjective and nuanced views.

5.3 Utilizing ordinal attributes

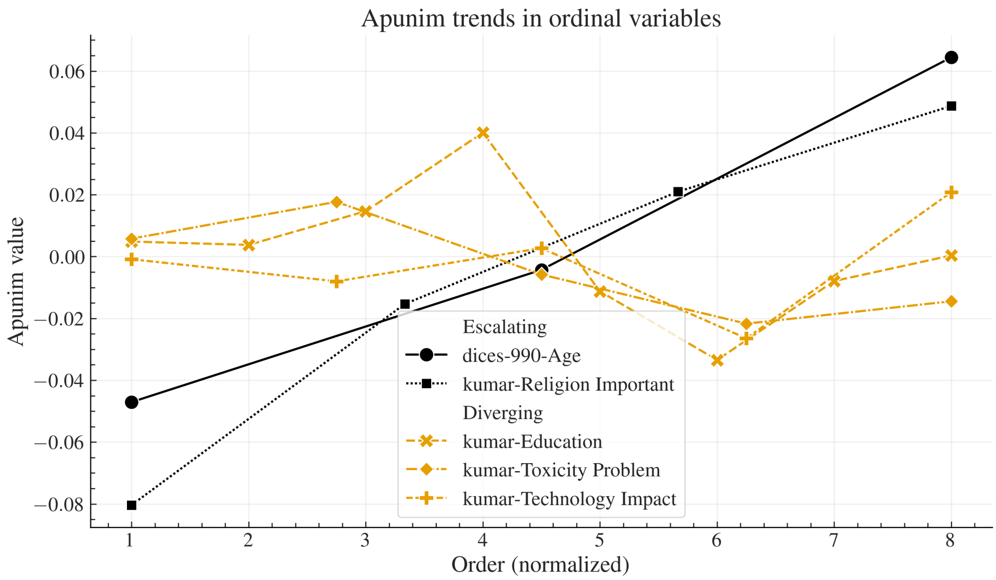

Fig. 7 The behavior of ordinal variables in all datasets with more than one statistically significant value. The x-axis represents the order of each factor (marked as `1), 2), ...` in Tables 2,3,5,4). The x-axis is normalized to the smallest common denominator between the number of ordinal values.

Analysis on annotated data frequently ignores the different kinds of measurements in studies using sociodemographic and LIKERT-style attributes. Researchers often treat all dimensions as *nominal/categorical* even though some are inherently ordinal (e.g., LIKERT-scale, education), and some continuous/ratio variables (e.g., age). Others attempt to find an "optimal point" which can be used to binarize continuous variables[12], but this approach is misguided since by using any artificial cut-off point, the resulting analysis is immediately invalidated by any change in the underlying data. Binarization also discards valuable information not present in binary or nominal categories By treating these variables as categorical, not only do we lose valuable information, but we may magnify statistical errors[20]. We posit that *retaining the order of ordinal attributes reveals interesting trends* that would be otherwise missed with categorical inference.

Figure 7 shows the trend lines for each ordinal PC attribute that has at least two statistically significant values. We aggregate these trends in two patterns; *"Clustering"* and *"Escalating"*.

Clustering

Some ordinal attributes tend to cluster around a few dominant values; for example, *Age* and *Education* in the Kumar dataset. This pattern is expected because the apunim algorithm is designed to identify clusters of annotations, causing annotations for nearby or similar values to be absorbed into the main effects driven by a few dominant modes.

Escalating

Age in DICES-990 and *Religion Importance* in the Kumar dataset follow a strict, monotonic trend. This implies that, in some cases, the *amount of polarization increases the further apart an opinion is from neutrality.* For example, non-religious annotators are more divided among themselves, while very religious annotators agree among themselves, but fundamentally disagree with other annotators.

6 How many annotators do we need?

While our test works best with many annotations per item, the cost required for multiple human annotators is often greatly restrictive [16]. It is thus imperative that we find the minimum number of annotators needed for *apunim* to generate stable and reliable estimates.

For each comment in all datasets, we randomly sample a subset of annotators with replacement and compute the pol_{obs} value (see Equation 3). We repeat the procedure 30 times (since larger values lead to large computation times—see Appendix D), and compute the standard error between the observations. We removed values where the standard error exploded due to a lack of comments annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 74 annotators—see Appendix B).

Figure 8 demonstrates the effect of the number of annotators on the reliability of the polarization estimation. Each dataset features a different standard error even with the same number of annotators, which is likely caused by the varying amount of comments used for each dataset. However, we observe a clear, non-linear, inverse

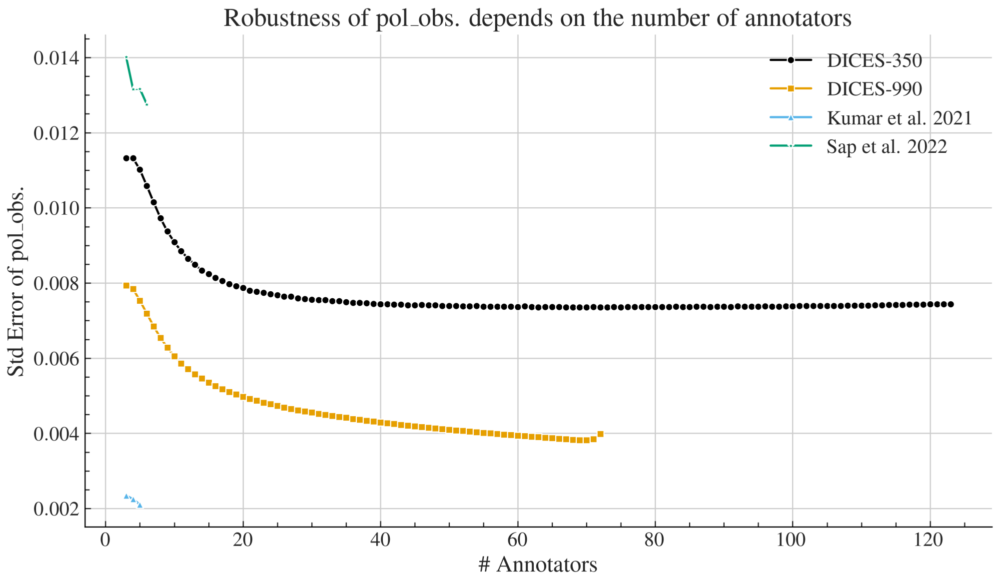

Fig. 8 Std error of the pol_{obs} statistic (Equation 3) when sampled with replacement for various number of annotators. We repeat the sampling process thirty times, and plot the average pol_{obs} value.

relationship between annotator count and standard error in all datasets. This can be attributed to a more robust histogram estimation used by nDFU (§3.3) when provided with more points for each of the subgroups. Furthermore, we can see that the metric becomes consistent with about 20 annotators per comment for the DICES datasets, after which we start to get diminishing results. Unfortunately, more datasets with large numbers of annotators that include data about their PC characteristics, are needed to generalize this finding.

7 Conclusion

We introduced the *apunim* metric which attributes polarization to annotator subgroups for various PC dimensions. Our metric works well for imbalanced annotator splits, is well-defined for multi-label annotation, includes a statistical significance test, and unlike previous work, quantifies the amount of attribution in order to allow for comparisons between various experiments and datasets. Using this metric, we discover that annotator ethnicity is the strongest and most consistent driver of polarization between annotators across datasets, and is much more pronounced on Hate Speech detection tasks. We also discover that education and religion consistently impact the behavior of annotators. Specifically, we discover that very religious annotators fundamentally agree with themselves but disagree with others in Toxicity detection tasks, a trend that is gradually and consistently diminished, and ultimately reversed, by irreligious annotators. We also provide an open-source, PyPi-installable implementation, and an estimation of the number of annotators needed for each , for our method to be consistently reliable.

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Appendix A Acronyms

LLM Large Language Model
PC Personal Characteristic
nDFU normalized Distance From Unimodality
FWER Family-Wise Error Rate
NLP Natural Language Processing
GPLv3 GNU General Public Licence v3

Appendix B Annotation details

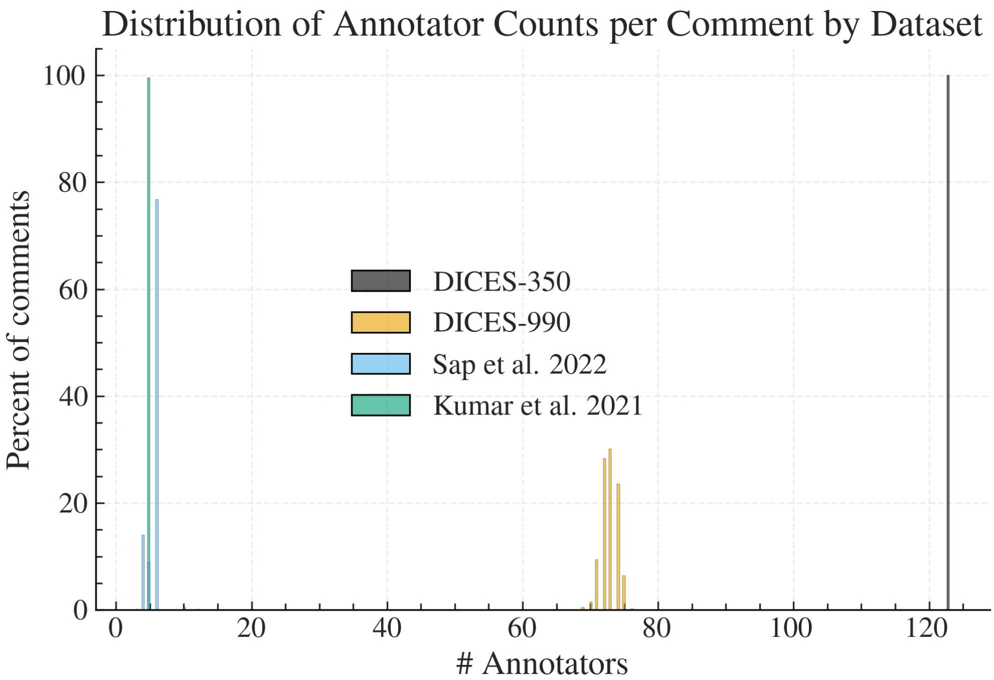

Fig. B1 We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table B1).

Table B1 Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	650
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12

The number of annotators per comment for all datasets presented in §4 can be found in Figure B1, and descriptive statistics can be found in Table B1. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 5, but not for the annotator number experiment in Figure 8 since they would produce uninterpretable figures.

Appendix C Software

C.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi (https://pypi.org/project/apunim). We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (https://dimits-ts.github.io/apunim).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (https://github.com/dimits-ts/apunim). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

C.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{P}_i$), which are originally calculated to obtain the apriori polarization values (see Equation 2), for the p-value calculations (§3.4). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{P}_i$ in Equation 2), and the number of apunim values used in the p-value calculation ($\tilde{D}_i$ in Algorithm 1) have to be the same.

Appendix D Replication Details

We use version `1.0.1` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 2) and the p-value estimation (§3.4), and set a FWER rate of 0.95.

The current version of the library handles large number of annotators very well due to the use of vectorized operations, but struggles when faced with a lot of comments when the annotators belong to multiple groups. Indeed, while the experiments for all datasets presented in §4 need no more than a minute to run, the Kumar dataset requires about 12 hours of computation even with a 20% sample. The annotator variance analysis presented in §6 also required more than 10 hours of computation, although that is expected due to the nature of the experiment, which is very computationally demanding, and a non-standard use of our library.

This bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package [21] for our own experiments).